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To cite this article: Jodavid A. Ferreira , Hemílio Coêlho & Abraão D. C. Nascimento (2021) A family of divergence-based classifiers for Polarimetric Synthetic Aperture Radar (PolSAR) imagery vector and matrix features, International Journal of Remote Sensing, 42:4, 1201-1229, DOI: [10.1080/01431161.2020.1826060](https://doi.org/10.1080/01431161.2020.1826060)

To link to this article: <https://doi.org/10.1080/01431161.2020.1826060>



Published online: 07 Dec 2020.



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A family of divergence-based classifiers for Polarimetric Synthetic Aperture Radar (PolSAR) imagery vector and matrix features

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ABSTRACT

Polarimetric Synthetic Aperture Radar (PolSAR) is one of the most important remote sensing tools. However, PolSAR images are strongly contaminated by a multidimensional interference (called speckle noise), making their processing (e.g. in the classification context) difficult. In terms of structure, multilook PolSAR data follow a definite positive hermitian behaviour and, therefore, require tailored classifiers for their features. Some classic classifiers – such as Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), K-Nearest Neighbours (KNN), and Support Vector Machine (SVM) – have been yielding unacceptable performance to these data, when applied directly. One justification is because they do not often take into account neither the speckle presence nor properties which are inherent to under-study relieves. This paper addresses a collection of PolSAR divergence-based classifiers, deduced from the normal, skew-normal, *t*-Student, and skew-*t* vector models as well as the scaled complex Wishart (SCW) distribution. The last model is a standard supposition to describe multilook PolSAR data, having two parameters: covariance matrix (which is directed to data nature) and number of looks (which controls the speckle noise effect). The considered remainder laws aim to model the main diagonal of these data, known as multivariate intensities. The performance of proposed methods is quantified and compared with those due to the Kullback-Leibler (KL) distance for multivariate normal distribution and to LDA, QDA, KNN, and SVM methods. Experiments with both artificial and real PolSAR data are considered. Results favour optimal Rényi classifiers for an Airborne Synthetic Aperture Radar (AIRSAR) image of San Francisco and the *t*-Student KL classifier for an SAR image system of the Electromagnetics Institute (EMISAR) image of Foulum.

ARTICLE HISTORY

Received 23 January 2020
Accepted 11 August 2020

1. Introduction

Polarimetric Synthetic Aperture Radar (PolSAR) has been indicated as an important and efficient remote sensing tool. This can be justified by three reasons at least: Such system can operate on all weather conditions, provide high-spatial resolution images, and furnish

several informations on a same scene. A PolSAR (which refers to the multilook version) scene can be understood as a map wherein each entry is associated with a positive definite hermitian matrix. This requires the use of multivariate processing methods on that support (Frery, Nascimento, and Cintra 2014). Besides, PolSAR images are strongly affected by a multidimensional interference pattern, called speckle noise. This effect imposes to PolSAR data that their real scalar features (known as intensities) do not follow a Gaussian behaviour and, as a consequence, tailored techniques need to be used. In particular, efficient classification tools are sought in PolSAR image processing.

Some works have addressed to propose classifiers for PolSAR data. From a literature review, one can note they can be categorized on three lines:

- (a) Polarimetric target decomposition: These methods were first formalized by Huynen (1970). After several classifiers have been proposed in the literature; e.g. Cloude (1992); Cloude and Pottier (1996, 1997); Lee et al. (1999); Ferro-Famil, Pottier, and Lee (2001); Touzi and Charbonneau (2002); Michaelsen, Soergel, and Thoennessen (2006); Touzi (2007); Licciardi et al. (2014).
- (b) Statistical models for PolSAR data: Many parametric, nonparametric, and Bayesian models jointly with their respective classifiers have been introduced in the PolSAR data literature:
 - (b₁) In the parametric approach: Borghys et al. (2006); Alberga, Satalino, and Staykova (2008); Zhang, Wu, and Wei (2009); Ersahin, Cumming, and Ward (2010); Huang et al. (2011); Salehi, Sahebi, and Maghsoudi (2014); Uhlmann and Kiranyaz (2014); Torres et al. (2014); Liu et al. (2015); Hou et al. (2015); Masjedi, Zoj, and Maghsoudi (2016); Xie, Jiao, and Zhao (2016).
 - (b₂) In the nonparametric approach: Dell'Acqua, Gamba, and Trianni (2006); Liu et al. (2016);
 - (b₃) In the Bayesian approach: Bouhlef and Méric (2019); Doulgeris, Anfinson, and Eltoft (2008).
- (c) Combination of (a) and (b), called *hybrid method*: Giacinto, Roli, and Bruzzone (2000); Hoekman, Vissers, and Tran (2011); Singh et al. (2013); Jagdhuber, Hajnsek, and Papathanassiou (2015); Zhang and Wu (2011); Salehi, Sahebi, and Maghsoudi (2014); Duan et al. (2017).

This paper presents advances at the branch (b).

Using multivariate analysis techniques directly has not shown to be a good way to classify PolSAR data. Mishra (2008) has presented evidence that linear discriminant analysis (LDA) applied to classify SAR multivariate intensities can produce inadequate performance. According to Tao et al. (2015), feature extraction methods combined with the K-Nearest Neighbours (KNN) and support vector machine classifiers can improve classification rates on PolSAR image processing. Borghys et al. (2006) have described a new feature-based supervised classification method for multichannel SAR data based on logistic regression (LR) and multinomial logistic regression (MNL) for separating different classes, taking into account spatial regularization. Results showed new methods may present meaningful improvement, comparatively to the maximum likelihood classifier. Giacinto, Roli, and Bruzzone (2000) reported the superiority of one algorithm (combined neural and statistical properties) for supervised classification of remote-sensing images.

Hoekman and Vissers (2003) proposed classifiers for heterogeneous PolSAR regions, where the backscatter behaviour was described by the complex Wishart distribution. Those classifiers were evaluated under two perspectives (separating and combining the C- and L-bands) and it was concluded that those in terms of combined bands presented the best performance. Assuming that PolSAR backscatters follow the K-Wishart distribution, Doulgeris, Anfinsen, and Eltoft (2007) proposed a Bayesian classification scheme, which can be used in both supervised and unsupervised approaches.

Stochastic distance is an important quantity in the Information Theory area, which was pioneered by Shannon (1948). He introduced three basic concepts: information, entropy, and relative entropy (or divergence). Based on the last, one defines distance between two probability measures. The first divergence concept was proposed by Kullback-Leibler (KL) and after several of classes of divergences were proposed. Divergence measures were subjected to a formal treatment by Csiszár (1967) and by Salicrú et al. (1994), resulting in divergence ϕ and $h - \phi$ classes, respectively. Some studies have addressed to derive the KL divergence for univariate and multivariate distributions. Contreras-Reyes and Arellano-Valle (2012) derived it for the multivariate skew-normal distribution and Contreras-Reyes (2014) proposed an asymptotic expression of the KL quantity between multivariate asymmetric heavy-tailed (in particular, t -Student and skew- t) distributions. Recently, the interest in adapting statistical information theory techniques to image processing contexts has grown (Nascimento, Cintra, and Frery 2010). Nascimento, Cintra, and Frery (2010) proposed eight divergence measures between $\mathcal{G}_l^0$ distributions for SAR intensity data. In this paper, we use stochastic distances derived from different probability models as main part of new classifiers.

Image processing techniques commonly work under the Gaussian assumption, due to its tractability. However, this condition is not often satisfied in practice. In particular, PolSAR data are represented by hermitian matrices whose elements of main diagonals are not described by the multivariate Gaussian distribution. In this paper, we propose a class of classifiers induced from information theory measures to assess the effect of non-Gaussianity when classifying PolSAR data. To that end, we consider KL and Jeffreys (J) divergences between Gaussian models compared with ones obtained from the multivariate t -Student, skew- t , and skew-normal distributions. We also consider the classification based on the Rényi divergence for the scaled complex Wishart (SCW) distribution. Results point out different classifiers for considered synthetic and real experiments, what motives the use of the proposed class.

This paper is organized as follows. Section 2 presents some models for PolSAR data. In Section 3, used divergences are discussed as particular cases into the $h - \phi$ class. The classifiers proposed in this paper are presented in Section 4. Subsequently, numerical results are given in Section 5. Finally, the main conclusions are furnished in Section 6.

2. Models for multilook Polarimetric SAR data

Full polarimetric SAR systems register complex scattering coefficients defined on four received and transmitted complex linear polarizations: Say S_{HH} (horizontal-horizontal), S_{HV} (horizontal-vertical), S_{VH} (vertical-horizontal), and S_{VV} (vertical-vertical). In practice, the assumption $S_{HV} = S_{VH}$ holds because the reciprocity theorem is satisfied (Ulaby and Elachi 1990). In general, it is common to assume systems having p polarization channels.

Let $y_i = (S_1^{(i)} S_2^{(i)} \dots S_p^{(i)})^T \in \mathbb{C}^p$, where $(\cdot)^T$ represents the transpose operator, for $i = 1, \dots, L$, be a random sample obtained from $y \sim \mathcal{N}_p^C(\mathbf{0}, \Sigma)$ having Σ as the hermitian covariance matrix. Thus, $\mathbf{Z} = L^{-1} \Sigma_{i=1}^L y_i y_i^*$ follows a scaled complex Wishart distribution (Frery, Nascimento, and Cintra 2014), where $(\cdot)^*$ represents the conjugate transpose operator. This situation is denoted by $\mathbf{Z} \sim \mathcal{W}_p^C(\Sigma, L)$ and this distribution satisfies $E\{\mathbf{Z}\} = \Sigma$ having probability density function (pdf) expressed by:

$$f_{\mathbf{Z}}(\dot{\mathbf{Z}}; \Sigma, L) = \frac{L^p |\dot{\mathbf{Z}}|^{L-p}}{|\Sigma|^L \Gamma_p(L)} \exp\left[-L \text{tr}(\Sigma^{-1} \dot{\mathbf{Z}})\right], \quad (1)$$

where $\dot{\mathbf{Z}}$ is an outcome of $\mathbf{Z}$, $\Gamma_p(L) = \pi^{p(p-1)/2} \prod_{i=0}^{p-1} \Gamma(L-i)$, for $L \geq p$, represents the multivariate gamma function, $\Gamma(\cdot)$ is the gamma function, and $\text{tr}(\cdot)$ is the trace operator. This model is the basic assumption for PolSAR data. The $\mathcal{W}_p^C$ model takes into account a control parameter for the effect of multidimensional speckle noise (say L) as well as Σ , which maps the dependence among the polarization channels. However, although this model has a pdf with tractable analytic form, it does not incorporate the effect of relief roughness. In this paper, we use the $\mathcal{W}_p^C$ model and other multivariate distributions for elaborating new classifiers. We also utilize an extension of (1) in order to construct more flexible synthetic scenarios, obtained from the multiplicative approach discussed as follows.

PolSAR data are often modelled by the multiplicative approach which says (see Freitas, Frery, and Correia 2004):

‘A pixel of a PolSAR image is described by random matrix $\mathbf{Z}_1 = X \cdot \mathbf{Z}$, where X is associated with the backscatter, $\mathbf{Z}$ describes the multidimensional speckle noise (following the SCW law), and X and $\mathbf{Z}$ are independent.’

Consider as model for the backscatter the reciprocal gamma distribution – termed by $X \sim \Gamma^{-1}(-\beta, (-\beta - 1)\mu)$ – with pdf given by:

$$f_X(x; \mu, \beta) = \frac{1}{\Gamma(-\beta)} \frac{1}{(-\beta - 1)\mu} \left[\frac{(-\beta - 1)\mu}{x} \right]^{\beta+1} \exp\left(-\frac{(-\beta - 1)\mu}{x}\right), \quad (2)$$

where $\mu > 0$ (scale) and $\beta < 0$ (shape) and x is a simple realization of X . The distribution for $\mathbf{Z}_1$ is presented as follows.

According to Freitas, Frery, and Correia (2004), making $\mu = 1$ in (2), the $\mathcal{G}_p^0$ distribution is obtained as a special case at the multiplicative model and its pdf is given by

$$f_{\mathbf{Z}_1}(\dot{\mathbf{Z}}; \beta, \Sigma, L) = \frac{L^p \Gamma(Lp - \beta)}{|\Sigma|^L (-\beta - 1)^\beta \Gamma(-\beta) \Gamma_p(L)} |\dot{\mathbf{Z}}|^{L-p} \left[(-\beta - 1) + L \text{tr}(\Sigma^{-1} \dot{\mathbf{Z}}) \right]^{\beta-Lp}, \quad (3)$$

where β is the roughness parameter. We use such model for generating synthetic scenarios on which will be quantified the performance of our proposals.

3. Stochastic distances

Divergence can be understood as a non-negative function between two probability measures which obey the property of definitiveness (see Frery, Nascimento, and Cintra 2014, for details). Let x and y be two random vectors defined on a same support $\mathcal{Z}$ having

densities $f_x(\cdot; \theta_1)$ and $f_y(\cdot; \theta_2)$, respectively, where θ_1 and θ_2 are parameters vectors. A class of divergence between f_x and f_y is defined by

$$D_\phi^h(x||y) = h \left[E_y \left(\phi \left(\frac{f_x(y; \theta_1)}{f_y(y; \theta_2)} \right) \right) \right], \quad (4)$$

where $E_y(\cdot)$ is the expected value of an argument with respect to a random vector y , $h : (0, \infty) \rightarrow [0, \infty)$ is a function strictly increasing, and $\phi : (0, \infty) \rightarrow [0, \infty)$ is a convex function such that $\phi(1) = h(0) = 0$ and $\phi''(1) > 0$. In additional, let $\mathbf{X}$ and $\mathbf{Y}$ be two random matrices defined on a same support $\mathcal{Z}$ equipped with densities $f_x(\cdot; \theta_1)$ and $f_y(\cdot; \theta_2)$, one has that the divergence between $\mathbf{X}$ and $\mathbf{Y}$ is given by

$$D_\phi^h(\mathbf{X}||\mathbf{Y}) = h \left[E_{\mathbf{Y}} \left(\phi \left(\frac{f_{\mathbf{X}}(\mathbf{Y}; \theta_1)}{f_{\mathbf{Y}}(\mathbf{Y}; \theta_2)} \right) \right) \right]. \quad (5)$$

In what follows, we specify (4) and (5) to obtain the KL, J, and Rényi divergences and distances:

$$[\phi(x), h(x)] = \begin{cases} \text{Divergences :} \\ \left[x \log(x), x \right] & , \text{ for KL,} \\ \left[(x-1) \log(x), x \right] & , \text{ for J,} \\ \left[\frac{x^a - a(x-1) - 1}{a-1}, \frac{1}{a-1} \log((a-1)x + 1) \right]_{a \in (0,1)} & , \text{ for Rényi.} \\ \text{Distances :} \\ \left[\frac{x^{1-a} + x^a - x - 1}{(a-1)}, \frac{1}{a-1} \log \left(\frac{(a-1)}{2} x + 1 \right) \right]_{a \in (0,1)} & , \text{ for Rényi.} \end{cases}$$

3.1. Used KL divergences

To the best of our knowledge, it is worth mentioning only Hagedorn et al. (2006) have derived the density of the joint distribution of multivariate intensities (extracted of a PolSAR return) from the nonscaled complex Wishart law, called trivariate chi-squared distribution. However, the density introduced by Hagedorn et al. (2006) is analytically intractable (among other difficulties, it is dependent of the first kind Bessel modified functions) for deducing information theory measures. Thus, we opt to use as probabilistic assumptions for multivariate SAR intensities (beyond the normal assumption)

- the t distribution for having tails heavier than the normal distribution keeping the symmetric and
- skew-normal and skew- t laws for covering symmetric and asymmetric phenomena.

Given two p -variate vectors x and y , Table 1 presents the used expressions for the Kullback-Leibler divergence of multivariate normal (from now on, we assume the terms 'Gaussian' and 'normal' as equivalent to keep the used laws origin notation), skew-normal (Contreras-Reyes and Arellano-Valle 2012; Contreras-Reyes 2014), multivariate t (Hofert 2013) and skew- t distributions (Contreras-Reyes 2014). The quantities in Table 1 will be computed by evaluating the maximum likelihood estimators (MLEs) under two samples their entries.

Table 1. Kullback-Leibler divergences for the multivariate normal, skew-normal multivariate, t -Student multivariate and skew- t multivariate distributions.

Model	KL Divergence
$x \sim N_p(\mu_1, \Sigma_1)$	$D_{KL}^N(\mu_1, \Sigma_1 \ \mu_2, \Sigma_2) = \frac{1}{2} \left\{ \log \left(\frac{ \Sigma_2 }{ \Sigma_1 } \right) + \text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_1 - \mu_2)^T \Sigma_2^{-1} (\mu_1 - \mu_2) - p \right\}.$
$y \sim N_p(\mu_2, \Sigma_2)$	
$x \sim SN_p(\mu_1, \Sigma_1, \eta_1)$	$D_{KL}^{SN}(\mu_1, \Sigma_1, \eta_1 \ \mu_2, \Sigma_2, \eta_2) = D_{KL}^N(\mu_1, \Sigma_1 \ \mu_2, \Sigma_2) + (7)$
$y \sim SN_p(\mu_2, \Sigma_2, \eta_2)$	$\frac{\sqrt{\delta_1}(\mu_1 - \mu_2)^T \delta_1}{\Sigma_2^{-1}} + E\{\log[2\Phi(W_{1,1})]\} - E\{\log[2\Phi(W_{2,1})]\}.$
where	$W_{1,1} \sim SN_1(0, \eta_1^T \Sigma_1 \eta_1, (\eta_1^T \Sigma_1 \eta_1)^{\frac{1}{2}}),$ $W_{2,1} \sim SN_1(\eta_2^T (\mu_1 - \mu_2), \eta_2^T \Sigma_1 \eta_2, \Xi)$ $\Xi = \eta_2^T \delta_1 [\eta_2^T \Sigma_1 \eta_2 - (\eta_2^T \delta_1)^2]^{-1/2}$
$x \sim t_p(\mu_1, \Sigma_1, v_1)$	$D_{KL}^t(\mu_1, \Sigma_1, v_1 \ \mu_2, \Sigma_2, v_2) = \frac{1}{2} \log\{(2\pi)^p \Sigma_2 \} + (8)$
$y \sim t_p(\mu_2, \Sigma_2, v_2)$	$\frac{1}{2} \left(\frac{v_2 + p}{v_2} \right) E[Q_2] - H(x_0) + \frac{1}{2} \log \Sigma_1 .$
$\psi(\cdot)$ is the digamma function	$H(x_0) = -\log\{B_p(v_1)\} + \left(\frac{v_1 + p}{2}\right) \left[\psi\left(\frac{v_1 + p}{2}\right) - \psi\left(\frac{v_1}{2}\right)\right],$ $B_p(v_1) = \left[\Gamma\left(\frac{v_1 + p}{2}\right)\right] / \left[\Gamma\left(\frac{v_1}{2}\right)(v_1 \pi)^{\frac{p}{2}}\right],$
where	$E[Q_2] = \frac{v_1}{(v_1 - 2)} \text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_1 - \mu_2)^T \Sigma_2^{-1} (\mu_1 - \mu_2)$
and	$x_0 \sim T_p(0, I_p, v_1).$
$x \sim St_p(\mu_1, \Sigma_1, \eta_1, v_1)$	$D_{KL}^{St}(\mu_1, \Sigma_1, \eta_1, v_1 \ \mu_2, \Sigma_2, \eta_2, v_2) = \frac{1}{2} \log\{(2\pi)^p \Sigma_2 \} (9)$
$y \sim St_p(\mu_2, \Sigma_2, \eta_2, v_2)$	$+ \frac{1}{2} \left(\frac{v_2 + p}{v_2} \right) E[Q_2] - E\left[\log\left\{2T\left(\frac{\eta_2^T}{\sqrt{\eta_2^T \Sigma_1 \eta_2 - (\eta_2^T \delta_1)^2}}(x - \mu_2); v_2 + p\right)\right\}\right]$ $- H(x_0) - E\left[\log\left\{2T\left(\frac{\sqrt{v_1 + p} \ \eta_1\ W}{\sqrt{v_1 + p - 1 + W^2}}; v_1 + p\right)\right\}\right],$
where	$E[Q_2] = \frac{v_1}{(v_1 - 2)} \text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_1 - \mu_2)^T \Sigma_2^{-1} (\mu_1 - \mu_2) +$ $2 \frac{\sqrt{\eta_1^T} \Gamma\left(\frac{v_1 - 1}{2}\right)}{\Gamma\left(\frac{v_1}{2}\right)} (\mu_1 - \mu_2)^T \Sigma_2^{-1} \delta_1.$ $T(\cdot)$ is the Student's t cumulative distribution function.

One can note that the Equation (7) presents two quantities $E[\log\{2\Phi(W_{1,1})\}]$ and $E[\log\{2\Phi(W_{2,1})\}]$, which do not have closed-form expressions. In order to overcome this gap, we use the Monte Carlo integration. We use the following algorithm proposed by Contreras-Reyes and Arellano-Valle (2012) to generate $W_{1,1}$ and $W_{2,1}$:

- $W_{1,1} \equiv 0 + \delta|U_0| + \mathbf{U}$, where $U_0 \sim N(0, 1)$ $\mathbf{U} \sim N_1(\mathbf{0}, (\eta_1^T \Sigma_1 \eta_1) - \delta^2)$ and $\delta = \frac{(\eta_1^T \Sigma_1 \eta_1)^{\frac{3}{2}}}{\sqrt{1 + (\eta_1^T \Sigma_1 \eta_1)^2}};$
- $W_{2,1} \equiv \eta_2^T (\mu_1 - \mu_2) + \delta|U_0| + \mathbf{U}$, where $U_0 \sim \tilde{N}(0, 1)$ $\mathbf{U} \sim \tilde{N}_1(\mathbf{0}, (\eta_2^T \Sigma_1 \eta_2) - \delta^2)$ and $\delta = \frac{(\eta_2^T \Sigma_1 \eta_2) \left(\frac{\eta_2^T \delta_1}{\sqrt{\eta_2^T \Sigma_1 \eta_2 - (\eta_2^T \delta_1)^2}} \right)}{\sqrt{1 + \eta_1^T (\eta_2^T \Sigma_1 \eta_2) \eta_2}} \text{ and } \eta = \frac{\eta_2^T \delta_1}{\sqrt{\eta_2^T \Sigma_1 \eta_2 - (\eta_2^T \delta_1)^2}}.$

In practice, the skewness η is estimated by maximizing the SN likelihood function (Arellano-Valle, Gómez, and Quintana 2004). As this optimization problem does not present closed-form solutions, we consider the Newton-Raphson iterative method: For $j = 1, \dots, p$ and $Z_{1,j}, \dots, Z_{n,j}$ as a random sample from the j th component of $z \sim SN_p(\mu, \Sigma, \eta)$, $Z_j \sim SN_1(\mu_j, \sigma_j^2, \eta_j)$, the MLE for η_j is

$$\hat{\eta}_j = \arg \max_{\eta_j \in \mathbb{R}} \left\{ -\frac{n}{2} \log \frac{\pi \sigma_j^2}{2} - \frac{1}{2\sigma_j^2} \sum_{i=1}^n (Z_{ij} - \mu_j)^2 + \sum_{i=1}^n \log \Phi \left[\frac{\eta_j (Z_{ij} - \mu_j)}{\sqrt{\sigma_j^2}} \right] \right\}. \quad (10)$$

The quantity (7) is measured by plugging MLEs in it.

Further, for reproducing the Equation (9), assume $W \sim ST_1(0, 1, \|\bar{\eta}_1\|, \nu_1 + p - 1)$, $\|\bar{\eta}_1\| = (\eta_1^T \Sigma_1 \eta_1)^{\frac{1}{2}}$, $\bar{\eta}_1 = \Sigma_1^{1/2} \eta_1$, $H(x_0)$ is the Shannon entropy of $x_0 \sim T_p(\mathbf{0}, \mathbf{I}_p, \nu_1)$ given by

$$H(x_0) = -\log\{B_p(\nu_1)\} + \left(\frac{\nu_1 + p}{2}\right) \left[\psi\left(\frac{\nu_1 + p}{2}\right) - \psi\left(\frac{\nu_1}{2}\right) \right] \quad \text{and} \quad (11)$$

$$B_p(\nu_1) = \frac{\Gamma\left(\frac{\nu_1 + p}{2}\right)}{\Gamma\left(\frac{\nu_1}{2}\right)(\nu_1 \pi)^{\frac{p}{2}}}, \quad \text{for } \nu_1 > 0. \quad (12)$$

Figure 1 illustrates the relation between $D_{\text{KL}}^{\text{SN}}$ and D_{KL}^{N} . It is noticeable that $D_{\text{KL}}^{\text{SN}}$ approaches D_{KL}^{N} when η tends to zero. The relation between D_{KL}^{t} and D_{KL}^{N} is presented in Figure 2. When ν is sufficiently large, D_{KL}^{t} approaches D_{KL}^{N} .

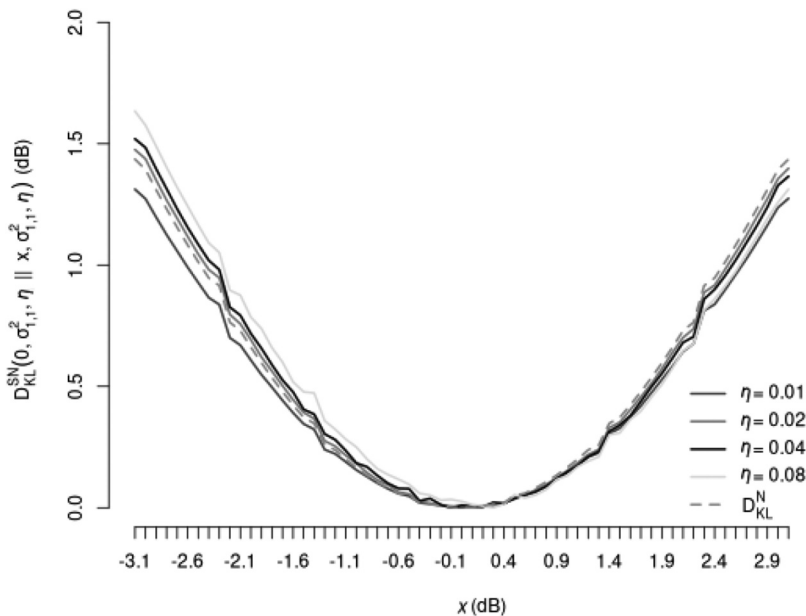


Figure 1. The KL divergence behaviour of the multivariate skew-normal distribution at different skewness coefficients. $D_{\text{KL}}^{\text{SN}}(0, \sigma_{1,1}^2, \eta || x, \sigma_{1,1}^2, \eta)$ is particular case univariate of the KL divergence for multivariate skew-normal distribution, $\sigma_{1,1}^2 = 3.36$, $x \in [-3.1, 3.1]$ and D_{KL}^{N} is the KL divergence for multivariate normal distribution.

3.2. Used Rényi divergences

First, we present a discussion about the used Rényi divergences for multivariate normal intensities. Let $\theta_1 = [\mu_1^T, \text{vec}(\Sigma_1)^T]^T$ and $\theta_2 = [\mu_2^T, \text{vec}(\Sigma_2)^T]^T$,

$$D_R^a(\theta_1 || \theta_2) = \frac{1}{a(a-1)} \log[K(\theta_1 || \theta_2; a)], \quad (13)$$

where $K(\theta_1 || \theta_2; a) = \int_{\mathbb{R}^p} f(z; \theta_1)^a f(z; \theta_2)^{1-a} dz$, $dz = \prod_{i=1}^p dz_i$, $\mu_i \in \mathbb{R}^p$, and $\Sigma_i \in \{Z \in \mathbb{R}^p \times \mathbb{R}^p : Z \succ 0\}$. According to Pardo (2005),

$$K(\theta_1 || \theta_2; a) = \frac{|a\Sigma_2 + (1-a)\Sigma_1|^{-\frac{1}{2}}}{|\Sigma_1|^{\frac{a-1}{2}} |\Sigma_2|^{-\frac{a}{2}}} \times \exp\left\{\frac{a(a-1)}{2} (\mu_1 - \mu_2)^T [a\Sigma_2 + (1-a)\Sigma_1]^{-1} (\mu_1 - \mu_2)\right\}, \quad (14)$$

and

$$D_R^a(\theta_1 || \theta_2) = \frac{1}{a(a-1)} \log[K(\theta_1 || \theta_2; a)] = (\mu_1 - \mu_2)^T [a\Sigma_2 + (1-a)\Sigma_1]^{-1} (\mu_1 - \mu_2) - \frac{1}{2a(a-1)} \log\left[\frac{|a\Sigma_2 + (1-a)\Sigma_1|}{|\Sigma_1|^{(1-a)} |\Sigma_2|^a}\right] \quad (15)$$

for $a \in \mathbb{R}$ and $a \neq 1$. In this study, we use four discrepancy measures based on the Rényi divergence:

$$\begin{cases} 1. & D_R^a(\theta_1 || \theta_2); \\ 2. & D_R^a(\theta_2 || \theta_1); \\ 3. & d_R^a(\theta_1 || \theta_2) = \frac{D_R^a(\theta_1 || \theta_2) + D_R^a(\theta_2 || \theta_1)}{2}; \\ 4. & d_{R_2}^a(\theta_1 || \theta_2) = \frac{1}{a(a-1)} \log\left[\frac{K(\theta_1 || \theta_2; a) + K(\theta_2 || \theta_1; a)}{2}\right]. \end{cases}$$

Second, we also use Rényi divergences for full PolSAR returns. Let $\theta_1 = [\text{vec}(\Sigma_1)^T, L_1]^T$ and $\theta_2 = [\text{vec}(\Sigma_2)^T, L_2]^T$ be parameter vectors of $\mathbf{Z}_1 \sim \mathcal{W}_p^C(\Sigma_1, L_1)$ and $\mathbf{Z}_2 \sim \mathcal{W}_p^C(\Sigma_2, L_2)$, where Σ_1 and Σ_2 are hermitian positive definite matrices and L_1 and L_2 are the numbers of looks. The Rényi divergences and distances are given by (Frery, Nascimento, and Cintra 2014):

$$\begin{cases} 1. & D_{R_C}^a(\theta_1 || \theta_2) = \frac{L}{a-1} \log\left[\frac{|(a\Sigma_1^{-1} + (1-a)\Sigma_2^{-1})^{-1}|}{|\Sigma_1|^a |\Sigma_2|^{1-a}}\right]; \\ 2. & D_{R_C}^a(\theta_2 || \theta_1) = \frac{L}{a-1} \log\left[\frac{|(a\Sigma_2^{-1} + (1-a)\Sigma_1^{-1})^{-1}|}{|\Sigma_2|^a |\Sigma_1|^{1-a}}\right]; \\ 3. & d_{R_C}^a(\theta_1 || \theta_2) = \frac{D_{R_C}^a(\theta_1 || \theta_2) + D_{R_C}^a(\theta_2 || \theta_1)}{2}; \\ 4. & d_{R_{2C}}^a(\theta_1 || \theta_2) = \frac{\log 2}{1-a} + \frac{1}{a-1} \times \log\left\{\left[\frac{|(a\Sigma_1^{-1} + (1-a)\Sigma_2^{-1})^{-1}|}{|\Sigma_1|^a |\Sigma_2|^{(1-a)}}\right]^L \right. \\ & \quad \left. + \left[\frac{|(a\Sigma_2^{-1} + (1-a)\Sigma_1^{-1})^{-1}|}{|\Sigma_2|^a |\Sigma_1|^{(1-a)}}\right]^L\right\}; \end{cases}$$

where $L_1 = L_2 = L$, by assuming the equivalent number of looks constant and known by the whole image.

4. Divergence-based classifiers for PolSAR data

Classifiers based on probability models are often used for PolSAR data (Gao, Yang, and Ma 2014; Moraes and Machado 2014; Cintra, Frery, and Nascimento 2013). One of the first classifiers in this sense was introduced by Lee, Grunes, and Kwok (1994), which is defined as follows: Let w_j be a class of PolSAR pixels with centroid $\mathbf{C}_j$ for $j = 1, \dots, k$ and k denote the number of populations in some context. The Lee's optimal Bayes classifier labels an entry $\mathbf{Z}_n$ with the class w_v if

$$d_w(\mathbf{Z}_n, w_v) = \inf_{j=1, \dots, v, \dots, k} d_w(\mathbf{Z}_n, w_j), \quad (16)$$

where $v \in \{1, \dots, k\}$ and

$$d_w(\mathbf{Z}_n, w_j) = \log |\mathbf{C}_j| + \text{tr}(\mathbf{C}_j^{-1} \mathbf{Z}_n). \quad (17)$$

We can say that $d_w(\cdot, \cdot)$ is as important for PolSAR data as the Mahalanobis distance (Bishop 2006, 80) is for classification over the Euclidean space. A measure which extends (17) is the KL divergence between SCW models, proposed by Frery, Nascimento, and Cintra (2014). Silva et al. (2013) proposed classifiers based on KL and Bhattacharyya distances for both multilook PolSAR and multivariate intensity data. In this section, we intend to develop more flexible divergence-based classifiers.

Silva et al. (2013) introduced and discussed classifiers for PolSAR segments by means of stochastic distances and hypothesis tests. One of those classifiers aimed to classify multivariate SAR images of intensities and optical sensor images when their associated data follow the normal multivariate model. In this paper, we extend some of methods proposed by Silva et al. (2013). To that end, we advance at constructing classifiers:

- (a) In terms of KL and J divergences for non-normal multivariate models;
- (b) Based on Rényi divergences for multivariate normal distributions;
- (c) With respect to the Rényi divergence for SCW laws.

For items (b) and (c), we consider optimum classifiers with respect to the order of the Rényi divergence. In what follows, we propose new classifiers. They are also compared with the classical LDA, QDA, KNN, and SVM classifiers (see James et al. 2014) as well as ones based on KL, Bhattacharyya, and J divergences between normal multivariate models, special cases in our proposals.

4.1. Training step

First assume that the support of PolSAR images may be partitioned in g segments, say $C_1, \dots, C_g$. PolSAR data in each segment are defined as

$$\left\{ \begin{array}{l} [z_i|C_k] = \begin{bmatrix} \|Z_i^{\text{Channel HH}}\|^2 \\ \|Z_i^{\text{Channel HV}}\|^2 \\ \|Z_i^{\text{Channel VV}}\|^2 \end{bmatrix} C_k, \text{ for multivariate intensities,} \\ \text{or} \\ [\mathbf{Z}_i|C_k] = \begin{bmatrix} \|Z_i^{\text{Channel HH}}\|^2 & Z_i^{\text{Channel HH HV}} & Z_i^{\text{Channel HH VV}} \\ & \|Z_i^{\text{Channel HV}}\|^2 & Z_i^{\text{Channel HV VV}} \\ & & \|Z_i^{\text{Channel VV}}\|^2 \end{bmatrix} C_k, \\ \text{for full PolSAR data,} \end{array} \right.$$

where $i = 1, \dots, N_k$, $k = 1, \dots, g$, and N_k is the number of pixel in the k th segment. In traditional classification problems, g samples are extracted from these segments called training samples. Under the hypothesis that (for $k = 1, \dots, g$), $[z_1|C_k], \dots, [z_{N_k}|C_k]$ represent a random sample obtained from $[z|C_k] \sim \{N_3(\mu_k, \Sigma_k), SN_3(\mu_k, \Sigma_k, \eta_k), t_3(\mu_k, \Sigma_k, \nu_k)\}$ or under the hypothesis that $[\mathbf{Z}_1|C_k], \dots, [\mathbf{Z}_{N_k}|C_k]$ are a random sample drawn from $[\mathbf{Z}|C_k] \sim \mathcal{W}_3^C(\Sigma_k, L_k)$, we fit vectorial and matrix models by the maximum likelihood estimation.

4.2. Divergence-based classifiers

Let $\hat{\theta}_k$ be ML estimates for θ_k , we have the following fitted multivariate distributions:

$$\hat{\theta}_k = \begin{cases} [\hat{\mu}_k^T, \text{vec}(\hat{\Sigma}_k)]^T, & \text{for multivariate normal distribution,} \\ [\hat{\mu}_k^T, \text{vec}(\hat{\Sigma}_k), \hat{\eta}_k^T]^T, & \text{for skew – normal multivariate distribution} \\ [\hat{\mu}_k^T, \text{vec}(\hat{\Sigma}_k), \hat{\nu}_k]^T, & \text{for } t \text{ – Student multivariate distribution,} \\ [\hat{\mu}_k^T, \text{vec}(\hat{\Sigma}_k), \hat{\eta}_k^T, \hat{\nu}_k]^T, & \text{for skew – } t \text{ multivariate distribution,} \\ [\text{vec}(\hat{\Sigma}_k), L]^T, & \text{for complex Wishart distribution with } L \text{ known.} \end{cases}$$

Now, consider we wish to test if a PolSAR return belongs to one among g clusters. Works about classification of PolSAR features assess this issue under two perspectives: (a) to extract test samples on which it is known their physical nature in order to quantify overall accuracy and to compare performance of distinct classifiers and (b) to test each entry of a PolSAR image to plot a classification map which is used as comparison qualitative measure. To that end, given a PolSAR return at the entry (h_1, h_2) , $\mathbf{Z}_h(h_1, h_2)$, define its neighbourhood for a window $u \times u$ with u odd as

$$\{ \mathbf{Z}_h(h_1 - u_1, h_2 - u_2) \quad \text{for } u_1, u_2 = -(u-1)/2, \dots, 0, \dots, (u-1)/2 \}$$

and, over it, obtain MLEs $\hat{\theta}_h$. We propose the class of classifiers:

$$\hat{i}_{\mathcal{M}} = \arg \min_{i \in \{1, \dots, g\}} \{d_{\mathcal{M}}(i)\} \quad (18)$$

where

$$d_{\mathcal{M}}(i) = \begin{cases} D_{\text{KL}}^{\text{N}}([\hat{\mu}_h^T, \text{vec}(\hat{\Sigma}_h)] || [\hat{\mu}_i^T, \text{vec}(\hat{\Sigma}_i)]), J^{\text{N}}(\cdot || \cdot), \\ D_{\text{KL}}^{\text{SN}}([\hat{\mu}_h^T, \text{vec}(\hat{\Sigma}_h), \hat{\eta}_h] || [\hat{\mu}_i^T, \text{vec}(\hat{\Sigma}_i), \hat{\eta}_i]), J^{\text{SN}}(\cdot || \cdot), \\ D_{\text{KL}}^{\text{t}}([\hat{\mu}_h^T, \text{vec}(\hat{\Sigma}_h), \hat{\nu}_h] || [\hat{\mu}_i^T, \text{vec}(\hat{\Sigma}_i), \hat{\nu}_i]), J^{\text{t}}(\cdot || \cdot), \\ D_{\text{KL}}^{\text{St}}([\hat{\mu}_h^T, \text{vec}(\hat{\Sigma}_h), \hat{\eta}_h^T, \hat{\nu}_h] || [\hat{\mu}_i^T, \text{vec}(\hat{\Sigma}_i), \hat{\eta}_i^T, \hat{\nu}_i]), J^{\text{St}}(\cdot || \cdot), \\ D_{\text{R}}^{\alpha}([\hat{\mu}_h^T, \text{vec}(\hat{\Sigma}_h)] || [\hat{\mu}_i^T, \text{vec}(\hat{\Sigma}_i)]), D_{\text{R}}^{\alpha}([\hat{\mu}_i^T, \text{vec}(\hat{\Sigma}_i)] || [\hat{\mu}_h^T, \text{vec}(\hat{\Sigma}_h)]), \\ d_{\text{R}}^{\alpha}([\hat{\mu}_h^T, \text{vec}(\hat{\Sigma}_h)], [\hat{\mu}_i^T, \text{vec}(\hat{\Sigma}_i)]), d_{\text{R}_2}^{\alpha}(\cdot, \cdot), \\ D_{\text{RC}}^{\alpha}([\text{vec}(\hat{\Sigma}_h), L] || [\text{vec}(\hat{\Sigma}_i), L]), D_{\text{RC}}^{\alpha}([\text{vec}(\hat{\Sigma}_i), L] || [\text{vec}(\hat{\Sigma}_h), L]), \\ d_{\text{RC}}^{\alpha}([\text{vec}(\hat{\Sigma}_h), L], [\text{vec}(\hat{\Sigma}_i), L]), d_{\text{R}_{2\text{C}}}^{\alpha}(\cdot, \cdot) \end{cases}$$

and $\mathcal{M} \in \{ \text{KLDN (KL divergence for normal), JDN (Jeffreys divergence for normal), KLDt (KL divergence for } t\text{-Student), JDt (Jeffreys divergence for } t\text{-Student), KLDSt (KL divergence for skew-normal), JDSN (Jeffreys divergence for skew-normal), KLDSt (KL divergence for skew-}t\text{), JDSt (Jeffreys divergence for skew-}t\text{), RDN1:}\alpha \text{ (Rényi divergence kind } (\theta_1 || \theta_2) \text{ for normal), RDN2:}\alpha \text{ (Rényi divergence kind } (\theta_2 || \theta_1) \text{ for normal), RdN1:}\alpha \text{ (Rényi first distance for normal), RdN2:}\alpha \text{ (Rényi second distance for normal), RDW1:}\alpha \text{ (Rényi divergence kind } (\theta_1 || \theta_2) \text{ for Wishart), RDW2:}\alpha \text{ (Rényi divergence kind } (\theta_2 || \theta_1) \text{ for Wishart), RdW1:}\alpha \text{ (Rényi first distance for Wishart) and RdW2:}\alpha \text{ (Rényi second distance for Wishart)} \}$. Thus, the class (18) includes sixteen new classifiers. The classifier RdW1: α has been proposed and applied successfully by Silva et al. (2013). The classification process is described in Algorithm 1. Some relations of the proposed classifiers are showed in the diagram of Figure 3.

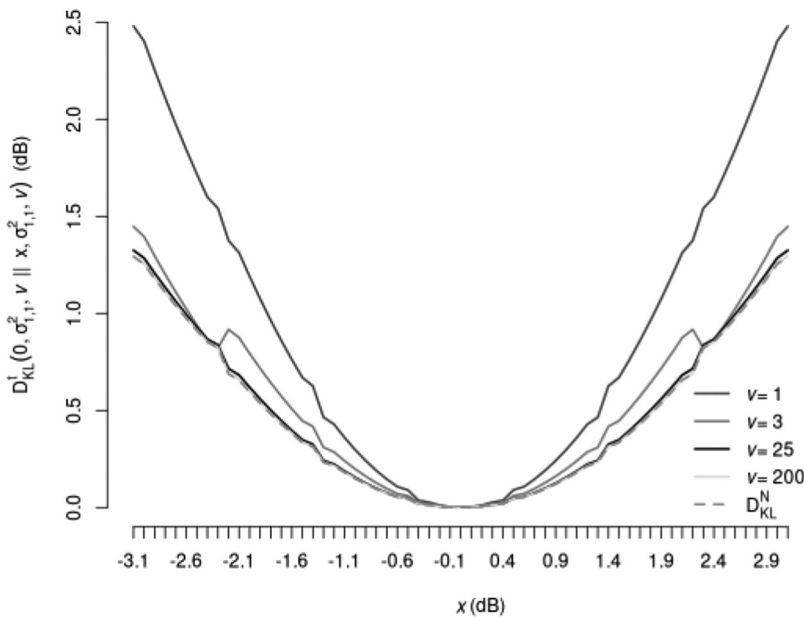


Figure 2. The KL divergence behaviour of the multivariate t distribution at different values of v . $D_{\text{KL}}^{\text{t}}(0, \sigma_{1,1}^2, v || x, \sigma_{1,1}^2, v)$ is particular case univariate of the KL divergence for multivariate t -Student distribution, $\sigma_{1,1}^2 = 3.36$, $x \in [-3.1, 3.1]$ and D_{KL}^{N} is the KL divergence for multivariate normal distribution.

Table 2. Classification matrix.

		Obtained classes ($\hat{c}$)			
		1	2	...	g
Actual classes (c)	1	$n_{1,1}$	$n_{1,2}$	...	$n_{1,g}$
	2	$n_{2,1}$	$n_{2,2}$	...	$n_{2,g}$
	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$n_{3,g}$
	g	$n_{g,1}$	$n_{g,2}$	$n_{g,3}$	$n_{g,g}$

where $\sum_{i=1}^g n_{i,i}$ is the sum of the main diagonal terms of the confusion matrix, $\sum_{i=1}^g \sum_{j=1}^g n_{i,j}$ is the sum of all its entries and g represents its pre-defined partitions.

Other used assessment measure is the Kappa coefficient (K) proposed by Cohen (1960). The Kappa coefficient is defined in terms of confusion matrix as:

$$K = \frac{P_0 - P_c}{1 - P_c}, \quad (20)$$

where $P_0 = N^{-1} \sum_{i=1}^g n_{i,i}$, $P_c = N^{-2} \sum_{i=1}^g n_{i,+} n_{+,i}$, $n_{i,i}$ is the sum of elements at the main diagonal of the confusion matrix, $n_{i,+}$ is the sum of its elements at i th row, $n_{+,i}$ is the sum of its elements at i th column and N represents the total number of decisions at the confusion matrix. Variance of Kappa (σ_K^2) was also calculated. According to Landis and Koch (1977), the Kappa coefficient may be interpreted by means of concordance percentage in Table 3.

5.1. Synthetic study

Our simulation study was organized as follows. $\mathcal{G}_p^0$ distributed data defined in (3) were generated such that three roughness levels were considered: $\beta = -1.5$ (representing arbitrarily extremely heterogeneous scenarios), $\beta = -6.0$ (for heterogeneous) and $\beta = -15.0$ (for homogeneous).

We also adopted three matrix parameters, which were defined from the San Francisco real image, discussed in detail subsequently. Figure 4 illustrates some kinds of target, which may be associated to the roughness parameter β . We assume the numbers of looks 3, 4 and 8 ($L = 3, 4$ and 8).

Figure 5(a) presents the used image signal and Figure 5(b,c) illustrate two generated images, having three different areas and, for each one, different roughnesses and data matrix. The legend in Figure 5(a) is defined by the following parametric configuration:

Table 3. Levels of correlation of Kappa coefficient.

Kappa statistics	Kind of concordance
< 0.00	Poor
0.00 to 0.20	Small
0.21 to 0.40	Standard
0.41 to 0.60	Moderate
0.61 to 0.80	Good
0.81 to 1.00	Excellent

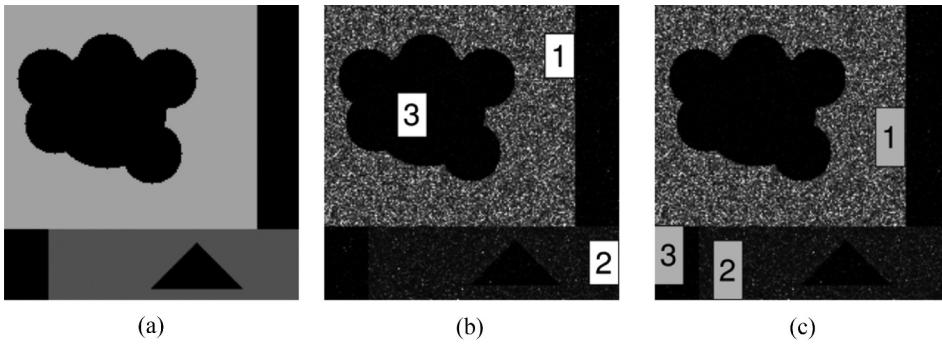


Figure 5. Synthetic image. (a) areas of synthetic image (the grey area is defined by $\beta = -15.0$ and Σ_A , the dark grey area is defined by $\beta = -6.0$ and Σ_B ; and the black area is defined by $\beta = -1.5$ and Σ_C); (b) training samples; (c) tests samples.

$$\begin{aligned}
 1) \beta &= -15.0 \text{ and } \Sigma_A = \begin{bmatrix} 0.385 & 0.118 + 0.008s & -0.098 - 0.009s \\ & 0.077 & -0.050 + 0.015s \\ & & 0.303 \end{bmatrix}, \left(\begin{array}{l} |\Sigma_A| = 4.1 \times 10^{-3}, \\ \text{tr}(\Sigma_A) = 0.765 \end{array} \right); \\
 2) \beta &= -6.0 \text{ and } \Sigma_B = \begin{bmatrix} 0.099 & 0.002 - 0.008s & -0.008 + 0.020s \\ & 0.044 & 0.001 + 0.002s \\ & & 0.096 \end{bmatrix}, \left(\begin{array}{l} |\Sigma_B| = 3.8 \times 10^{-4}, \\ \text{tr}(\Sigma_B) = 0.238 \end{array} \right); \\
 3) \beta &= -1.5 \text{ and } \Sigma_C = \begin{bmatrix} 0.008 & 0.001 - 0.001s & 0.011 + 0.002s \\ & 0.001 & -0.000 + 0.002s \\ & & 0.025 \end{bmatrix}, \left(\begin{array}{l} |\Sigma_C| = 5.22 \times 10^{-8}, \\ \text{tr}(\Sigma_C) = 0.034 \end{array} \right);
 \end{aligned}$$

where $s = \sqrt{-1}$ represents the imaginary unit.

Selected training samples are highlighted in Figure 5(b), while test areas samples are displayed in Figure 5(c). For each simulation result, it was used 100 Monte Carlo replicas and number of looks three. Furthermore, results were obtained varying window sizes as 3×3 , 5×5 , and 7×7 .

In Table 4, results of accuracy percentage for twenty studied classifiers are quantified when the number of looks is three. For the window 3×3 , the best classifications were obtained by the classifiers equipped with the Rényi divergences based on the SCW distribution (RDW1: α and RDW2: α). In this case, RDW1:0.1 and RDW2:0.9 presented $A(\%) = 98.30\%$.

For windows 5×5 and 7×7 , the two previous classifiers obtained also the highest values of $A(\%)$. The worst classification was obtained by the Jeffreys divergence based on the multivariate skew- t distribution (JDSt), with accuracy of 31.16% at the window 7×7 . Problems with estimates have precluded the associated classifier. The KL divergence (KLDN) presented $A(\%)$ of 95.19% with window 3×3 . KNN, LDA, QDA (Quadratic Discriminant Analysis), and SVM (Support Vector Machine) yielded $A(\%)$ values 87.38%, 76.30%, 86.49%, and 90.45% respectively.

Table 5 displays simulation results for $L = 4$. The best classifier was RDW1: α at the 3×3 window and $\alpha = 0.1$, having accuracy of 98.30%. Using the symmetric divergence furnished the same success percentage, considering α to 0.9. The worst classifier was the

Table 4. Classification results for looks = 3.

	Window 3×3		Window 5×5		Window 7×7	
	α	A (%)	α	A (%)	α	A (%)
L3						
$D_R^a(\theta_1 \theta_2)$	0.7	96.70	0.6	96.96	0.6	97.08
$D_R^a(\theta_2 \theta_1)$	0.3	96.70	0.4	96.96	0.4	97.08
$d_R^a(\theta_1, \theta_2)$	0.5	95.10	0.5	96.13	0.5	96.43
$d_{R_2}^a(\theta_1, \theta_2)$	(0.3,0.7)	96.67	(0.4,0.6)	97.02	(0.4,0.6)	97.20
$D_{RC}^a(\theta_1 \theta_2)$	0.1	98.30	0.1	97.81	0.1	97.73
$D_{RC}^a(\theta_2 \theta_1)$	0.9	98.30	0.9	97.81	0.9	97.73
$d_{RC}^a(\theta_1, \theta_2)$	0.5	98.11	0.5	97.37	0.5	96.57
$d_{R_2C}^a(\theta_1, \theta_2)$	0.5	98.11	0.5	97.37	0.5	96.57
$D_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	95.19	-	95.01	-	94.20
$J_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	93.75	-	93.77	-	93.11
$D_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	95.19	-	95.01	-	94.20
$J_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	93.75	-	93.77	-	93.11
$D_{KL}^t([\mu_1, \Sigma_1, v_1] [\mu_2, \Sigma_2, v_2])$	-	73.95	-	74.85	-	75.27
$J^t([\mu_1, \Sigma_1, v_1] [\mu_2, \Sigma_2, v_2])$	-	63.99	-	64.78	-	65.08
$D_{KL}^{St}(\theta_1 \theta_2)$	-	32.05	-	31.78	-	31.38
$J^{St}(\theta_1 \theta_2)$	-	31.87	-	31.61	-	31.16
KNN				87.38		
LDA				76.30		
QDA				86.49		
SVM				90.45		
L3 – Box-Cox transformation						
$D_R^a(\theta_1 \theta_2)$	0.5	98.49	0.6	98.95	0.7	99.35
$D_R^a(\theta_2 \theta_1)$	0.5	98.49	0.4	98.95	0.3	99.35
$d_R^a(\theta_1, \theta_2)$	(0.5,0.6,0.7)	98.46	0.5	98.94	0.5	99.29
$d_{R_2}^a(\theta_1, \theta_2)$	0.5	98.49	(0.4,0.6)	98.95	(0.3,0.7)	99.35
$D_{RC}^a(\theta_1 \theta_2)$	-	-	-	-	-	-
$D_{RC}^a(\theta_2 \theta_1)$	-	-	-	-	-	-
$d_{RC}^a(\theta_1, \theta_2)$	-	-	-	-	-	-
$d_{R_2C}^a(\theta_1, \theta_2)$	-	-	-	-	-	-
$D_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	98.25	-	98.26	-	98.80
$J_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	98.05	-	98.40	-	98.87
$D_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	98.25	-	98.26	-	98.80
$J_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	98.05	-	98.40	-	98.87
$D_{KL}^t([\mu_1, \Sigma_1, v_1] [\mu_2, \Sigma_2, v_2])$	-	98.24	-	97.66	-	97.60
$J^t([\mu_1, \Sigma_1, v_1] [\mu_2, \Sigma_2, v_2])$	-	52.54	-	51.11	-	50.79
$D_{KL}^{St}(\theta_1 \theta_2)$	-	54.74	-	54.63	-	55.30
$J^{St}(\theta_1 \theta_2)$	-	56.79	-	56.79	-	57.03
KNN				88.67		
LDA				90.31		
QDA				90.70		
SVM				90.30		

Kullback-Leibler divergence based on the multivariate skew- t distribution (KLDSt) with 5×5 window. The KLDSt percentage was 28.13%.

For the 7×7 window, with exception of (i) KNN, LDA, QDA and (ii) KLDt, JDSt, KLDSt and JDSt, thirteen classifiers assumed accuracy value above of 90%.

It is known the increasing of number of looks leads more homogeneous scenarios. Table 6 approaches results when the number of looks is eight. The best rating percentages were obtained at the 3×3 window for RDW1: α with $\alpha = 0.1$ and RDW2: α with $\alpha = 0.9$ having percentage 98.26%.

Table 5. Classification results for looks = 4.

	Window 3 × 3		Window 5 × 5		Window 7 × 7	
	<i>a</i>	<i>A</i> (%)	<i>a</i>	<i>A</i> (%)	<i>a</i>	<i>A</i> (%)
L4						
$D_R^a(\theta_1 \theta_2)$	0.6	97.77	0.5	97.69	0.5	97.87
$D_R^a(\theta_2 \theta_1)$	0.4	97.77	0.5	97.69	0.5	97.87
$d_R^a(\theta_1, \theta_2)$	0.5	97.15	0.5	97.69	0.5	97.87
$d_{R_2}^a(\theta_1, \theta_2)$	(0.4,0.6)	97.77	(0.4,0.6)	97.84	0.5	97.87
$D_{RC}^a(\theta_1 \theta_2)$	0.1	98.30	0.1	97.75	0.1	97.67
$D_{RC}^a(\theta_2 \theta_1)$	0.9	98.30	0.9	97.75	0.9	97.67
$d_{RC}^a(\theta_1, \theta_2)$	0.5	98.12	(0.1,0.9)	97.42	(0.1,0.9)	96.62
$d_{R_2C}^a(\theta_1, \theta_2)$	(0.2,0.8)	98.13	(0.1,0.9)	97.42	(0.1,0.9)	96.62
$D_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	96.10	-	95.62	-	94.77
$J_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	93.81	-	93.69	-	93.04
$D_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	96.10	-	95.62	-	94.77
$J_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	93.81	-	93.69	-	93.04
$D_{KL}^t([\mu_1, \Sigma_1, \nu_1] [\mu_2, \Sigma_2, \nu_2])$	-	76.24	-	77.00	-	77.25
$J^t([\mu_1, \Sigma_1, \nu_1] [\mu_2, \Sigma_2, \nu_2])$	-	65.64	-	66.27	-	66.44
$D_{KL}^{St}(\theta_1 \theta_2)$	-	28.30	-	28.13	-	28.24
$J^{St}(\theta_1 \theta_2)$	-	28.44	-	28.34	-	28.49
KNN				89.57		
LDA				78.07		
QDA				88.75		
SVM				91.89		
L4 – Box-Cox transformation						
$D_R^a(\theta_1 \theta_2)$	0.5	98.65	0.6	99.04	0.7	99.34
$D_R^a(\theta_2 \theta_1)$	0.5	98.65	0.4	99.04	0.3	99.34
$d_R^a(\theta_1, \theta_2)$	0.5	98.65	(0.4,0.6)	99.04	0.5	99.32
$d_{R_2}^a(\theta_1, \theta_2)$	0.5	98.65	0.5	99.04	(0.3,0.7)	99.34
$D_{RC}^a(\theta_1 \theta_2)$	-	-	-	-	-	-
$D_{RC}^a(\theta_2 \theta_1)$	-	-	-	-	-	-
$d_{RC}^a(\theta_1, \theta_2)$	-	-	-	-	-	-
$d_{R_2C}^a(\theta_1, \theta_2)$	-	-	-	-	-	-
$D_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	98.27	-	98.24	-	98.76
$J_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	98.13	-	98.37	-	98.85
$D_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	98.27	-	98.24	-	98.76
$J_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	98.13	-	98.37	-	98.85
$D_{KL}^t([\mu_1, \Sigma_1, \nu_1] [\mu_2, \Sigma_2, \nu_2])$	-	98.27	-	97.87	-	98.30
$J^t([\mu_1, \Sigma_1, \nu_1] [\mu_2, \Sigma_2, \nu_2])$	-	79.88	-	79.95	-	80.27
$D_{KL}^{St}(\theta_1 \theta_2)$	-	54.54	-	54.74	-	54.77
$J^{St}(\theta_1 \theta_2)$	-	63.22	-	63.68	-	63.61
KNN				90.85		
LDA				91.70		
QDA				92.12		
SVM				91.89		

The Rényi divergences and distances based on the Gaussian distribution performed worse than Rényi divergences and distances based on the Wishart for the 5 × 5 and 7 × 7 windows. The classifiers based on the Rényi measure had better performance than the Kullback-Leibler and Jeffreys. The worst classifiers were the Kullback-Leibler and Jeffreys measures for the multivariate *t*-Student and multivariate skew-*t* distributions, where the JDSt classifier having the worst *A* (%) (31.16%).

Figure 5 displays the simulation reference map to compare maps of classifiers. The best classification (made by RDW1:0.1 or RDW2:0.9) is displayed in Figure 6(a). The worst results in Figure 6 were obtained by JDSt, but there were improvements for more

Table 6. Classification results for looks = 8.

	Window 3×3		Window 5×5		Window 7×7	
	α	$A(\%)$	α	$A(\%)$	α	$A(\%)$
L8						
$D_R^a(\theta_1 \theta_2)$	0.5	97.69	0.5	97.69	0.5	97.74
$D_R^a(\theta_2 \theta_1)$	0.5	97.69	0.5	97.69	0.5	97.74
$d_R^a(\theta_1, \theta_2)$	0.5	97.69	(0.4,0.6)	97.69	(0.3,0.7)	97.76
$d_{R_2}^a(\theta_1, \theta_2)$	0.5	97.69	0.5	97.69	0.5	97.74
$D_{RC}^a(\theta_1 \theta_2)$	0.1	98.26	0.1	97.65	0.1	97.65
$D_{RC}^a(\theta_2 \theta_1)$	0.9	98.26	0.9	97.65	0.9	97.65
$d_{RC}^a(\theta_1, \theta_2)$	(0.3,0.7)	98.12	(0.2,0.8)	97.41	(0.2,0.8)	97.41
$d_{R_2C}^a(\theta_1, \theta_2)$	(0.3,0.7)	98.13	(0.2,0.8)	97.41	(0.2,0.8)	97.41
$D_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	96.92	-	96.09	-	95.22
$J_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	96.12	-	95.51	-	94.81
$D_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	96.91	-	96.12	-	95.24
$J_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	95.45	-	94.90	-	94.16
$D_{KL}^t([\mu_1, \Sigma_1, v_1] [\mu_2, \Sigma_2, v_2])$	-	75.40	-	75.95	-	76.06
$J^t([\mu_1, \Sigma_1, v_1] [\mu_2, \Sigma_2, v_2])$	-	68.97	-	69.43	-	69.48
$D_{KL}^{St}(\theta_1 \theta_2)$	-	31.54	-	31.85	-	32.27
$J^{St}(\theta_1 \theta_2)$	-	32.37	-	32.75	-	33.14
KNN				93.72		
LDA				82.42		
QDA				93.04		
QDA				95.00		
L8 – Box-Cox transformation						
$D_R^a(\theta_1 \theta_2)$	0.7	98.89	0.8	99.44	0.8	99.69
$D_R^a(\theta_2 \theta_1)$	0.3	98.89	0.2	99.44	0.2	99.69
$d_R^a(\theta_1, \theta_2)$	0.5	98.79	0.5	99.12	0.5	99.28
$d_{R_2}^a(\theta_1, \theta_2)$	(0.3,0.7)	98.89	(0.2,0.8)	99.44	(0.2,0.8)	99.69
$D_{RC}^a(\theta_1 \theta_2)$	-	-	-	-	-	-
$D_{RC}^a(\theta_2 \theta_1)$	-	-	-	-	-	-
$d_{RC}^a(\theta_1, \theta_2)$	-	-	-	-	-	-
$d_{R_2C}^a(\theta_1, \theta_2)$	-	-	-	-	-	-
$D_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	98.47	-	98.85	-	99.20
$J_{KL}^N([\mu_1, \Sigma_1] [\mu_2, \Sigma_2])$	-	98.39	-	98.88	-	99.17
$D_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	98.47	-	98.85	-	99.21
$J_{KL}^{SN}([\mu_1, \Sigma_1, \eta_1] [\mu_2, \Sigma_2, \eta_2])$	-	98.39	-	98.88	-	99.17
$D_{KL}^t([\mu_1, \Sigma_1, v_1] [\mu_2, \Sigma_2, v_2])$	-	98.34	-	97.97	-	98.32
$J^t([\mu_1, \Sigma_1, v_1] [\mu_2, \Sigma_2, v_2])$	-	72.51	-	71.82	-	71.64
$D_{KL}^{St}(\theta_1 \theta_2)$	-	65.33	-	65.15	-	65.05
$J^{St}(\theta_1 \theta_2)$	-	65.74	-	64.50	-	63.96
KNN	94.37					
LDA				94.12		
QDA				95.27		
SVM				95.03		

homogeneous scenarios (for higher number of looks). The use of KLDN in multivariate intensities yielded good results, but it imposed difficult to classify textures of intermediate degrees. LDA, QDA, KNN, and SVM were affected by multidimensional speckle noise, but these results were improved when increasing the number of looks. In all cases, using the Box-cox transformation led improved results.

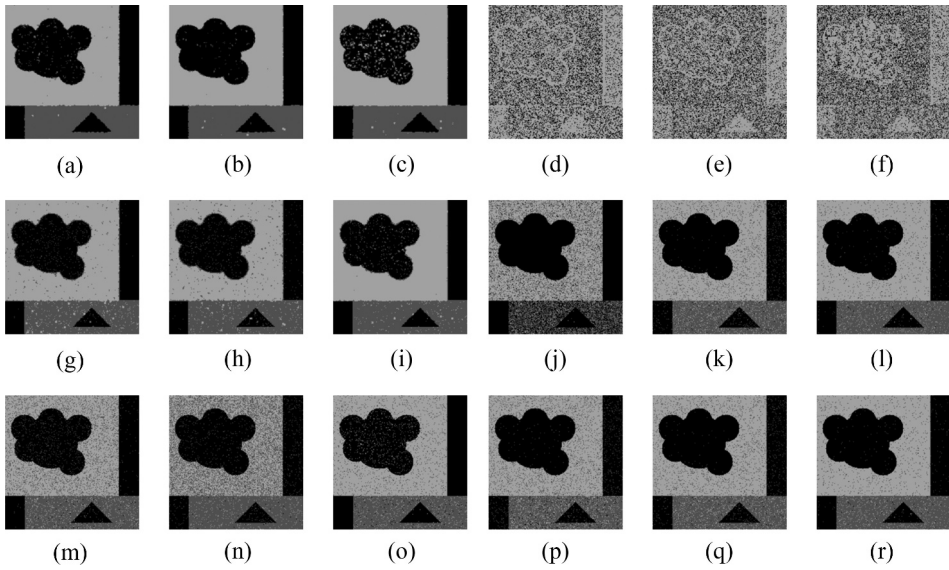


Figure 6. Classification of synthetic images for $L = 3, 4$ and 8 . (a) $L = 3$ and $D_{RC}^{0.1}(\theta_1||\theta_2) - 3 \times 3$; (b) $L = 4$ and $D_{RC}^{0.1}(\theta_1||\theta_2) - 3 \times 3$; (c) $L = 8$ and $D_{RC}^{0.1}(\theta_1||\theta_2) - 3 \times 3$; (d) $L = 3$ and $J^{St}(\theta_1||\theta_2) - 7 \times 7$; (e) $L = 4$ and $D_{KL}^{St}(\theta_1||\theta_2) - 5 \times 5$; (f) $L = 8$ and $D_{KL}^{St}(\theta_1||\theta_2) - 3 \times 3$; (g) $L = 3$ and $D_{KL}^N(\theta_1||\theta_2) - 3 \times 3$; (h) $L = 4$ and $D_{KL}^N(\theta_1||\theta_2) - 3 \times 3$; (i) $L = 8$ and $D_{KL}^N(\theta_1||\theta_2) - 3 \times 3$; (j) $L = 3$ and LDA; (k) $L = 4$ and LDA; (l) $L = 8$ and LDA; (m) $L = 3$ and QDA; (n) $L = 4$ and QDA; (o) $L = 8$ and QDA; (p) $L = 3$ and KNN; (q) $L = 4$ and KNN; (r) $L = 8$ and KNN.

5.2. Experiments with real data

In this section the proposed family of classifiers is applied to two real PolSAR images. The first experiment was made on a region of San Francisco bay (USA). This image was captured by the AIRSAR (Airborne Synthetic Aperture Radar) sensor with the number of looks four. Further, the second experiment was performed on a Foullum (DK) region. Resulting image was obtained by the EMISAR (SAR image system of the Electromagnetics Institute) sensor with number of look eight.

Figure 7 shows the AIRSAR image with highlighted training and test samples. In this case, we collect three regions for both training and test stages. They are chosen such that ones represent only one texture; i.e., sea (case less textured), forest (middle textured) and urban (strangely textured) scenarios. One the other hand, Figure 8 presents the EMISAR image, which has six well-defined regions (1 – Wheat, 2 – Rape, 3 – Rye, 4 – Conifer, 5 – Oat and 6 – background). The training and test samples are shaded.

Before discussing about classification results, we perform a brief multivariate descriptive analysis of data, obtained from both actual images. To that end, we consider a reduction of dimensionality by representing considered images by their two sample main components. Figures 7(c–d) and 8(c–d) display dispersion plots of such components. For the San Francisco image, note that data from the three regions are overlapped, that impose hard classification scenarios. Applying the Box-Cox transformation to data, one can note ocean regions become more distinct of forest and urban areas, but the last two ones present some degree of overlap. For the Foullum image, although one can segment

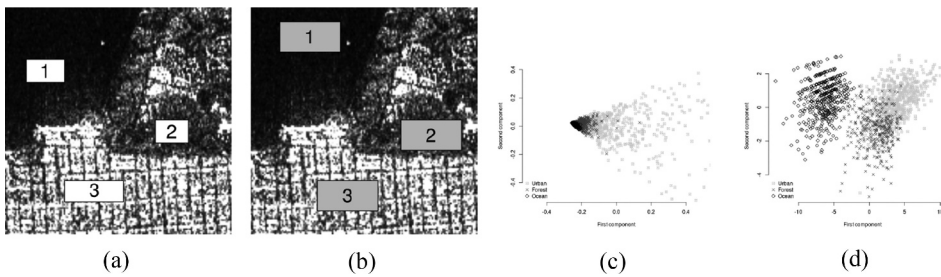


Figure 7. Operations on the San Francisco image. (a) training samples; (b) tests samples; (c) principal component analysis for real data; (d) principal component analysis for data transformed by Box-Cox.

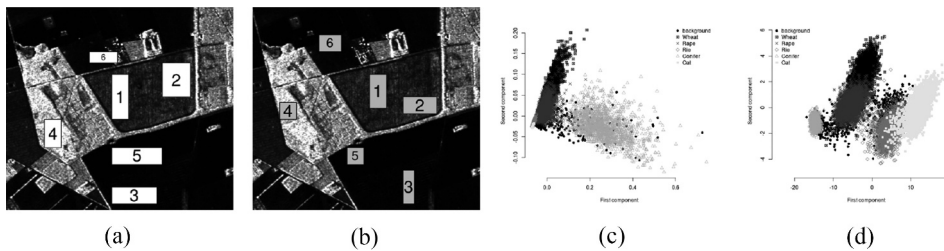


Figure 8. Operations on the Foulum image. (a) training samples; (b) tests samples; (c) principal component analysis for real data; (d) principal component analysis for data transformed by Box-Cox.

some distinct textures (such that background and conifer regions), there is expressive overlapping on both original and transformed data. Moreover, the last image involve more segments to be classified and, therefore, its classification seems to impose more difficulty than the first.

Results for the San Francisco image classification can be checked in the [Table 7](#). We use classifiers in two contexts, with original and Box-Cox transformation based Box and Cox (1964). As discussed in [Figures 7\(c–d\)](#) and [8\(c,d\)](#), one can see that the classification rates improved for transformed data. It can be explained by the fact of all multivariate distributions (with exception of the SCW model) we considered for obtaining the used distances are symmetric. There is not among the proposals the element that outperforms all the others. As follows, we present and discuss the better proposals.

The best rate for the real data was obtained with $A(\%)$ 99.69% by $RDN1:\alpha$ and $RDN2:\alpha$ with α of 0.9 and then its symmetric counterpart with α of 0.1, at window 7×7 . Its corresponding image can be found in [Figure 9\(b\)](#), in comparison to the San Francisco real image in [Figure 8](#). For the transformed data, the best classification furnished $A(\%)$ 99.73% and has been obtained through three proposed classifiers: $RDN1:0.9$, $RDN2:0.1$, and $RdN2:\alpha$ with α 0.1 and 0.9. The worst classifications was given by the JDSt with percentage of 38.06% and window 3×3 . For transformed data, the worst result was obtained by the same divergence at the window 5×5 .

Now, consider the Foulum image. The results can be found in [Table 8](#), in which the KLDt obtained the best classification at the window 5×5 , with accuracy 93.18%. The

Table 7. Classification results for San Francisco – USA.

	Window 3 × 3			Window 5 × 5			Window 7 × 7					
	α	A (%)	K	σ_k^2	α	A (%)	K	σ_k^2	α	A (%)	K	σ_k^2
San Francisco												
$D_R^g(\theta_2 \theta_1)$	0.9	96.24	94.37	3.14×10^{-5}	0.9	98.92	98.37	9.34×10^{-6}	0.9	99.69	99.54	2.69×10^{-6}
$d_R^g(\theta_1, \theta_2)$	0.1	96.24	94.37	3.14×10^{-5}	0.1	98.92	98.37	9.34×10^{-6}	0.1	99.69	99.54	2.69×10^{-6}
$d_R^g(\theta_1, \theta_2)$	0.5	83.46	75.20	1.14×10^{-4}	0.5	85.91	78.86	1.01×10^{-4}	0.5	87.14	80.72	9.41×10^{-5}
$d_R^g(\theta_1, \theta_2)$	(0.1,0.9)	94.89	92.33	4.21×10^{-5}	(0.1,0.9)	97.60	96.40	2.04×10^{-5}	(0.1,0.9)	98.06	97.10	1.65×10^{-5}
$D_R^g(\theta_1 \theta_2)$	0.9	91.09	86.64	6.96×10^{-5}	0.9	96.48	94.71	2.95×10^{-5}	0.9	98.72	98.08	1.10×10^{-5}
$D_R^g(\theta_2 \theta_1)$	0.1	91.09	86.64	6.96×10^{-5}	0.1	96.48	94.71	2.95×10^{-5}	0.1	98.72	98.08	1.10×10^{-5}
$d_R^g(\theta_1, \theta_2)$	(0.4,0.5,0.6)	87.81	81.71	9.08×10^{-5}	...[1]	94.54	91.81	4.47×10^{-5}	...[2]	97.44	96.17	2.16×10^{-5}
$d_R^g(\theta_1, \theta_2)$	...[3]	87.84	81.76	9.05×10^{-5}	...[4]	94.54	91.81	4.47×10^{-5}	...[5]	97.40	96.11	2.19×10^{-5}
$D_{KL}^N((\mu_1, \Sigma_1) (\mu_2, \Sigma_2))$	-	96.86	95.30	2.64×10^{-5}	-	98.64	97.97	1.16×10^{-5}	-	98.95	98.43	9.01×10^{-6}
$J^N((\mu_1, \Sigma_1) (\mu_2, \Sigma_2))$	-	87.46	81.18	9.36×10^{-5}	-	90.59	85.89	7.34×10^{-5}	-	93.19	89.78	5.51×10^{-5}
$D_{KL}^N((\mu_1, \Sigma_1, \eta_1) (\mu_2, \Sigma_2, \eta_2))$	-	96.86	95.30	2.64×10^{-5}	-	98.64	97.97	1.16×10^{-5}	-	98.95	98.43	9.01×10^{-6}
$J^N((\mu_1, \Sigma_1, \eta_1) (\mu_2, \Sigma_2, \eta_2))$	-	87.46	81.18	9.36×10^{-5}	-	90.59	85.89	7.34×10^{-5}	-	93.19	89.78	5.51×10^{-5}
$D_{KL}^S((\mu_1, \Sigma_1, v_1) (\mu_2, \Sigma_2, v_2))$	-	91.99	87.98	6.38×10^{-5}	-	94.46	91.69	4.54×10^{-5}	-	94.19	91.29	4.75×10^{-5}
$J^S((\mu_1, \Sigma_1, v_1) (\mu_2, \Sigma_2, v_2))$	-	58.08	31.13	2.71×10^{-4}	-	55.21	32.81	2.70×10^{-4}	-	55.48	33.22	2.63×10^{-4}
$D_{KL}^S(\theta_1 \theta_2)$	-	38.37	7.55	3.51×10^{-4}	-	40.03	10.05	3.27×10^{-4}	-	39.61	9.41	3.41×10^{-4}
$J^S(\theta_1 \theta_2)$	-	38.06	7.08	3.56×10^{-4}	-	39.45	9.17	3.33×10^{-4}	-	38.83	8.25	3.49×10^{-4}
KNN	A(%) →		89.78		K →		84.67		$\sigma_k^2 \rightarrow$		7.97×10^{-5}	
LDA	A(%) →		57.07		K →		35.60		$\sigma_k^2 \rightarrow$		2.30×10^{-4}	
QDA	A(%) →		85.33		K →		77.99		$\sigma_k^2 \rightarrow$		1.07×10^{-4}	
SVM	A(%) →		90.71		K →		83.86		$\sigma_k^2 \rightarrow$		8.34×10^{-5}	
San Francisco – Box-Cox transformation												
$D_R^g(\theta_1 \theta_2)$	0.8	99.07	98.60	8.01×10^{-6}	0.9	99.57	99.36	3.69×10^{-6}	0.9	99.73	99.59	2.35×10^{-6}
$D_R^g(\theta_2 \theta_1)$	0.2	99.07	98.60	8.01×10^{-6}	0.1	99.57	99.36	3.69×10^{-6}	0.1	99.73	99.59	2.35×10^{-6}
$d_R^g(\theta_1, \theta_2)$	0.5	98.64	97.96	1.16×10^{-5}	0.5	99.10	98.66	7.68×10^{-6}	0.5	99.38	99.07	5.36×10^{-6}
$d_R^g(\theta_1, \theta_2)$	(0.2,0.8)	99.07	98.60	8.01×10^{-6}	...[6]	99.54	98.66	7.68×10^{-6}	(0.1,0.9)	99.73	99.59	2.35×10^{-6}
$D_R^g(\theta_1 \theta_2)$	-	-	-	-	-	-	-	-	-	-	-	-
$D_R^g(\theta_2 \theta_1)$	-	-	-	-	-	-	-	-	-	-	-	-
$d_R^g(\theta_1, \theta_2)$	-	-	-	-	-	-	-	-	-	-	-	-
$d_R^g(\theta_1, \theta_2)$	-	-	-	-	-	-	-	-	-	-	-	-
$D_{KL}^N((\mu_1, \Sigma_1) (\mu_2, \Sigma_2))$	-	98.53	97.79	1.26×10^{-5}	-	99.19	98.78	7.02×10^{-6}	-	99.26	98.89	6.36×10^{-6}
$J^N((\mu_1, \Sigma_1) (\mu_2, \Sigma_2))$	-	97.60	96.39	2.04×10^{-5}	-	98.72	98.08	1.09×10^{-5}	-	99.07	98.60	8.01×10^{-6}
$D_{KL}^N((\mu_1, \Sigma_1, \eta_1) (\mu_2, \Sigma_2, \eta_2))$	-	98.53	97.79	1.26×10^{-5}	-	99.19	98.78	7.02×10^{-6}	-	99.26	98.89	6.36×10^{-6}

(Continued)

Table 7. (Continued).

	Window 3 × 3				Window 5 × 5				Window 7 × 7			
	a	$A(\%)$	K	σ_K^2	a	$A(\%)$	K	σ_K^2	a	$A(\%)$	K	σ_K^2
$J^{SN}([\mu_1, \Sigma_1, n_1][[\mu_2, \Sigma_2, n_2]])$	–	97.60	96.39	2.04×10^{-5}	–	98.72	98.08	1.09×10^{-5}	–	99.07	98.60	8.01×10^{-6}
$D_{KL}^*([\mu_1, \Sigma_1, v_1][[\mu_2, \Sigma_2, v_2]])$	–	98.14	97.21	1.58×10^{-5}	–	99.15	98.72	7.35×10^{-6}	–	99.15	98.72	7.35×10^{-6}
$J^S([\mu_1, \Sigma_1, v_1][[\mu_2, \Sigma_2, v_2]])$	–	98.22	97.32	1.52×10^{-5}	–	98.80	98.19	1.03×10^{-5}	–	98.99	98.49	8.67×10^{-6}
$D_{KL}^*(\theta_1 \parallel \theta_2)$	–	71.89	57.83	1.67×10^{-4}	–	70.11	55.16	1.74×10^{-4}	–	70.65	55.98	1.73×10^{-4}
$J^{St}(\theta_1 \parallel \theta_2)$	–	68.99	53.48	1.72×10^{-4}	–	67.87	51.80	1.77×10^{-4}	–	68.18	52.26	1.75×10^{-4}
KNN	$A(\%) \rightarrow$		90.44		$K \rightarrow$		85.65		$\sigma_K^2 \rightarrow$		7.51×10^{-5}	
LDA	$A(\%) \rightarrow$		88.73		$K \rightarrow$		83.10		$\sigma_K^2 \rightarrow$		8.65×10^{-5}	
QDA	$A(\%) \rightarrow$		87.77		$K \rightarrow$		81.64		$\sigma_K^2 \rightarrow$		9.31×10^{-5}	
SVM	$A(\%) \rightarrow$		90.71		$K \rightarrow$		86.06		$\sigma_K^2 \rightarrow$		7.33×10^{-5}	

Table 8. Classification results for Foulum – DK.

	Window 3 × 3				Window 5 × 5				Window 7 × 7			
	α	$A(\%)$	K	σ_K^2	α	$A(\%)$	K	σ_K^2	α	$A(\%)$	K	σ_K^2
Foulum												
$D_R^a(\theta_1 \theta_2)$	0.7	81.67	77.65	2.50×10^{-5}	0.8	83.47	79.83	2.31×10^{-5}	0.9	84.42	80.98	2.22×10^{-5}
$D_R^a(\theta_2 \theta_1)$	0.3	81.67	77.65	2.50×10^{-5}	0.2	83.47	79.83	2.31×10^{-5}	0.1	84.42	80.98	2.22×10^{-5}
$d_R^a(\theta_1, \theta_2)$	0.5	81.43	77.38	2.51×10^{-5}	(0.3,0.7)	82.61	78.81	2.37×10^{-5}	(0.4,0.6)	83.30	79.65	2.30×10^{-5}
$d_R^a(\theta_1, \theta_2)$	0.5	81.43	77.38	2.51×10^{-5}	(0.2,0.8)	82.66	78.87	2.38×10^{-5}	...[1]	83.31	79.67	2.30×10^{-5}
$D_R^a(\theta_1 \theta_2)$	0.8	70.22	63.91	3.59×10^{-5}	0.8	71.72	65.70	3.47×10^{-5}	0.8	71.51	65.43	3.46×10^{-5}
$D_R^a(\theta_2 \theta_1)$	0.2	70.22	63.91	3.59×10^{-5}	0.2	71.72	65.70	3.47×10^{-5}	0.2	71.51	65.43	3.46×10^{-5}
$d_R^a(\theta_1, \theta_2)$	(0.2,0.8)	69.94	63.40	3.60×10^{-5}	(0.2,0.8)	71.00	64.61	3.55×10^{-5}	(0.2,0.8)	70.36	63.78	3.58×10^{-5}
$d_R^a(\theta_1, \theta_2)$	(0.2,0.8)	67.31	60.26	3.72×10^{-5}	(0.2,0.8)	67.58	60.45	3.70×10^{-5}	(0.2,0.8)	67.20	59.96	3.68×10^{-5}
$D_{R,C}^N(\mu_1, \Sigma_1 [\mu_2, \Sigma_2])$	-	80.23	75.86	2.69×10^{-5}	-	82.69	78.86	2.43×10^{-5}	-	83.52	79.86	2.34×10^{-5}
$J^N([\mu_1, \Sigma_1 [\mu_2, \Sigma_2])$	-	75.92	70.64	3.11×10^{-5}	-	79.30	74.75	2.79×10^{-5}	-	80.23	75.88	2.69×10^{-5}
$D_{KL}^{SN}(\mu_1, \Sigma_1, \eta_1 [\mu_2, \Sigma_2, \eta_2])$	-	80.23	75.86	2.69×10^{-5}	-	82.69	78.86	2.43×10^{-5}	-	83.52	79.86	2.34×10^{-5}
$J^{SN}(\mu_1, \Sigma_1, \eta_1 [\mu_2, \Sigma_2, \eta_2])$	-	75.92	70.64	3.11×10^{-5}	-	79.30	74.75	2.79×10^{-5}	-	80.23	75.88	2.69×10^{-5}
$D_{KL}^L([\mu_1, \Sigma_1, v_1 [\mu_2, \Sigma_2, v_2])$	-	90.67	88.60	1.51×10^{-5}	-	93.18	91.67	1.14×10^{-5}	-	93.05	91.50	1.16×10^{-5}
$J^L(\mu_1, \Sigma_1, v_1 [\mu_2, \Sigma_2, v_2])$	-	79.34	74.87	2.89×10^{-5}	-	82.13	78.24	2.60×10^{-5}	-	82.99	79.30	2.50×10^{-5}
$D_{KL}^{Rv}(\theta_1 \theta_2)$	-	27.58	12.36	3.60×10^{-5}	-	28.78	13.78	3.67×10^{-5}	-	29.05	14.16	3.69×10^{-5}
$J^{Rv}(\theta_1 \theta_2)$	-	30.66	16.43	3.81×10^{-5}	-	30.86	16.60	3.81×10^{-5}	-	31.73	17.68	3.85×10^{-5}
KNN	$A(\%) \rightarrow$		85.04		$K \rightarrow$		81.77		$\sigma_K^2 \rightarrow$		2.26×10^{-5}	
LDA	$A(\%) \rightarrow$		70.22		$K \rightarrow$		62.2		$\sigma_K^2 \rightarrow$		3.68×10^{-5}	
QDA	$A(\%) \rightarrow$		74.54		$K \rightarrow$		71.01		$\sigma_K^2 \rightarrow$		3.09×10^{-5}	
SVM	$A(\%) \rightarrow$		86.53		$K \rightarrow$		83.59		$\sigma_K^2 \rightarrow$		2.06×10^{-5}	
Foulum – Box-Cox transformation												
$D_R^a(\theta_1 \theta_2)$	0.9	92.10	90.38	1.29×10^{-5}	0.9	93.67	92.28	1.05×10^{-5}	0.9	94.79	93.65	8.78×10^{-6}
$D_R^a(\theta_2 \theta_1)$	0.1	92.10	90.38	1.29×10^{-5}	0.1	93.67	92.28	1.05×10^{-5}	0.1	94.79	93.65	8.78×10^{-6}
$d_R^a(\theta_1, \theta_2)$	0.5	90.41	88.31	1.52×10^{-5}	0.5	91.25	89.35	1.39×10^{-5}	0.5	91.97	90.22	1.29×10^{-5}
$d_R^a(\theta_1, \theta_2)$	(0.1,0.9)	92.07	90.34	1.29×10^{-5}	(0.1,0.9)	93.64	92.25	1.05×10^{-5}	(0.1,0.9)	94.69	93.54	8.93×10^{-6}
$D_R^a(\theta_1 \theta_2)$	-	-	-	-	-	-	-	-	-	-	-	-
$D_R^a(\theta_2 \theta_1)$	-	-	-	-	-	-	-	-	-	-	-	-
$d_R^a(\theta_1, \theta_2)$	-	-	-	-	-	-	-	-	-	-	-	-
$d_R^a(\theta_1, \theta_2)$	-	-	-	-	-	-	-	-	-	-	-	-
$D_{R,C}^N([\mu_1, \Sigma_1 [\mu_2, \Sigma_2])$	-	92.08	90.35	1.29×10^{-5}	-	94.33	93.09	9.53×10^{-6}	-	94.90	93.79	8.63×10^{-6}

(Continued)

Table 8. (Continued).

	Window 3 × 3				Window 5 × 5				Window 7 × 7			
	<i>a</i>	<i>A</i> (%)	<i>K</i>	σ_K^2	<i>a</i>	<i>A</i> (%)	<i>K</i>	σ_K^2	<i>a</i>	<i>A</i> (%)	<i>K</i>	σ_K^2
$J^N(\mu_1, \Sigma_1 \mu_2, \Sigma_2)$	–	86.46	83.50	2.01×10^{-5}	–	88.29	85.73	1.78×10^{-5}	–	88.75	86.29	1.72×10^{-5}
$D_{KL}^{SN}(\mu_1, \Sigma_1, \eta_1 \mu_2, \Sigma_2, \eta_2)$	–	92.08	90.35	1.29×10^{-5}	–	94.33	93.09	9.53×10^{-6}	–	94.90	93.79	8.63×10^{-6}
$J^{SN}(\mu_1, \Sigma_1, \eta_1 \mu_2, \Sigma_2, \eta_2)$	–	86.46	83.50	2.01×10^{-5}	–	88.29	85.73	1.78×10^{-5}	–	88.75	86.29	1.72×10^{-5}
$D_{KL}^E(\mu_1, \Sigma_1, v_1 \mu_2, \Sigma_2, v_2)$	–	93.99	92.68	1.01×10^{-5}	–	95.95	95.07	6.96×10^{-6}	–	96.77	96.06	5.61×10^{-6}
$J^E(\mu_1, \Sigma_1, v_1 \mu_2, \Sigma_2, v_2)$	–	68.98	62.16	3.65×10^{-5}	–	68.68	61.81	3.64×10^{-5}	–	68.19	61.23	3.67×10^{-5}
$D_{KL}^{ST}(\theta_1 \theta_2)$	–	78.14	73.29	2.90×10^{-5}	–	78.38	73.58	2.88×10^{-5}	–	78.32	73.52	2.88×10^{-5}
$J^{ST}(\theta_1 \theta_2)$	–	75.64	70.21	3.12×10^{-5}	–	76.13	70.81	3.08×10^{-5}	–	76.76	71.59	3.02×10^{-5}
KNN	$A(\%) \rightarrow$		86.37		$K \rightarrow$		83.40		$\sigma_K^2 \rightarrow$		2.08×10^{-5}	
LDA	$A(\%) \rightarrow$		86.26		$K \rightarrow$		83.29		$\sigma_K^2 \rightarrow$		2.09×10^{-5}	
QDA	$A(\%) \rightarrow$		85.53		$K \rightarrow$		82.37		$\sigma_K^2 \rightarrow$		2.20×10^{-5}	
SVM	$A(\%) \rightarrow$		87.22		$K \rightarrow$		84.43		$\sigma_K^2 \rightarrow$		1.97×10^{-5}	

same classifier presented the best results for transformed data at the window 7×7 , with $A(\%)$ 96.77%. Classified images by KLDt on both original and transformed data are exhibited in Figure 10(b,c), respectively. The worst classifications for no transformed data were obtained by KLDSt at window 3×3 with accuracy of 27.58%; for to

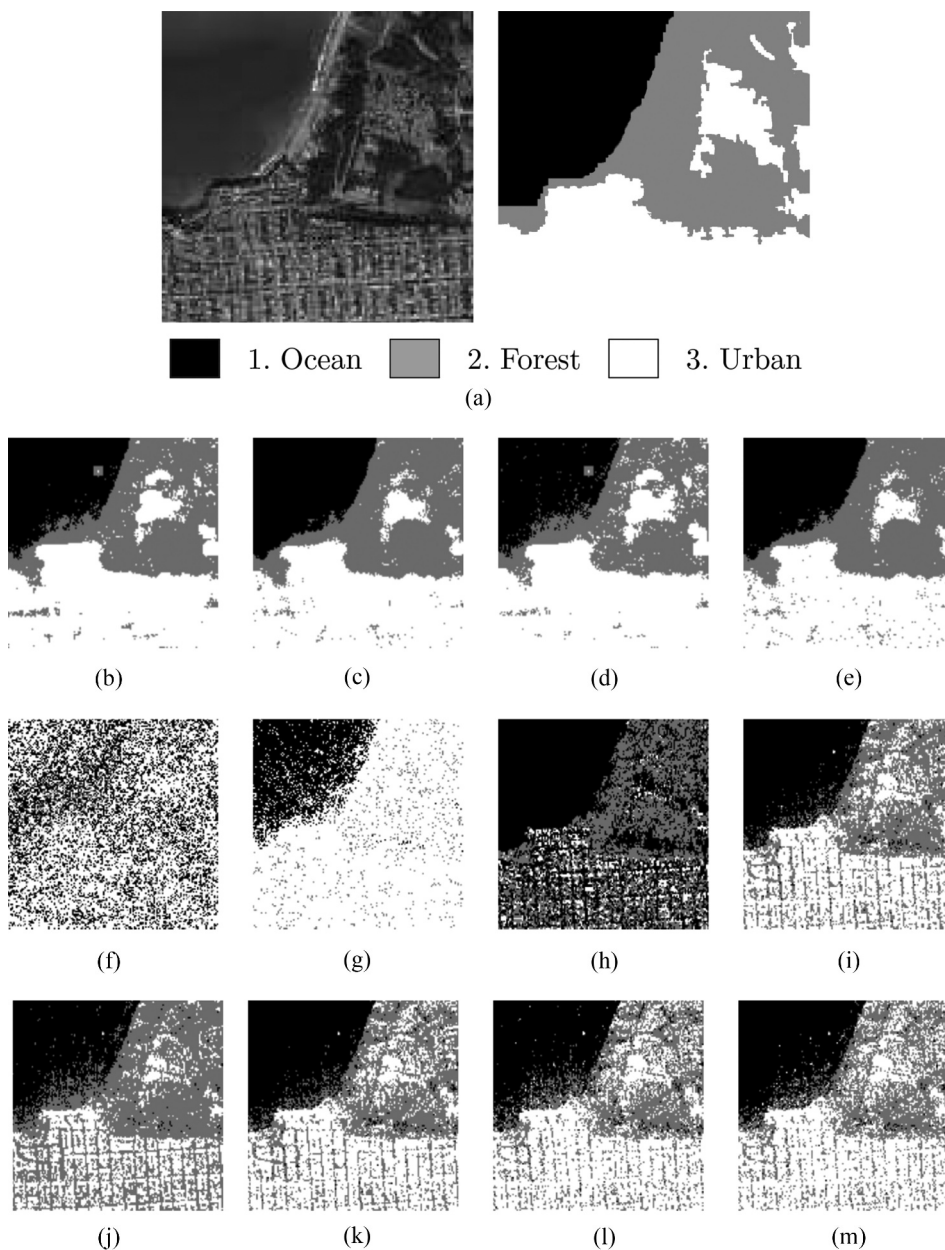


Figure 9. Classification results on the San Francisco AIRSAR image. (a) Ground-truth data of San Francisco; (b) $D_R^{0.9}(\theta_1||\theta_2) - 7 \times 7$; (c) $D_R^{0.9}(\theta_1||\theta_2) - 7 \times 7$ - Box-Cox; (d) $D_{KL}^N(\theta_1||\theta_2) - 7 \times 7$; (e) $D_{KL}^N(\theta_1||\theta_2) - 7 \times 7$ - Box-Cox; (f) $J^{St}(\theta_1||\theta_2) - 3 \times 3$; (g) $J^{St}(\theta_1||\theta_2) - 5 \times 5$ - Box-Cox; (h) LDA; (i) LDA - Box-Cox; (j) QDA (k) QDA - Box-Cox; (l) KNN; (m) KNN - Box-Cox.

transformed data, the JDt at window 7×7 showed the worst classification with accuracy 68.19%. The related images are on Figure 10(f,g).

Now, we discuss shortly results due to classic measures. The KL divergences obtained performance smaller than 90% at the three windows for original data, and, for transformed data, the classification performance was 94.90% at the window 7×7 . The classifiers LDA, QDA, KNN and SVM performed better for transformed data than for the original versions.

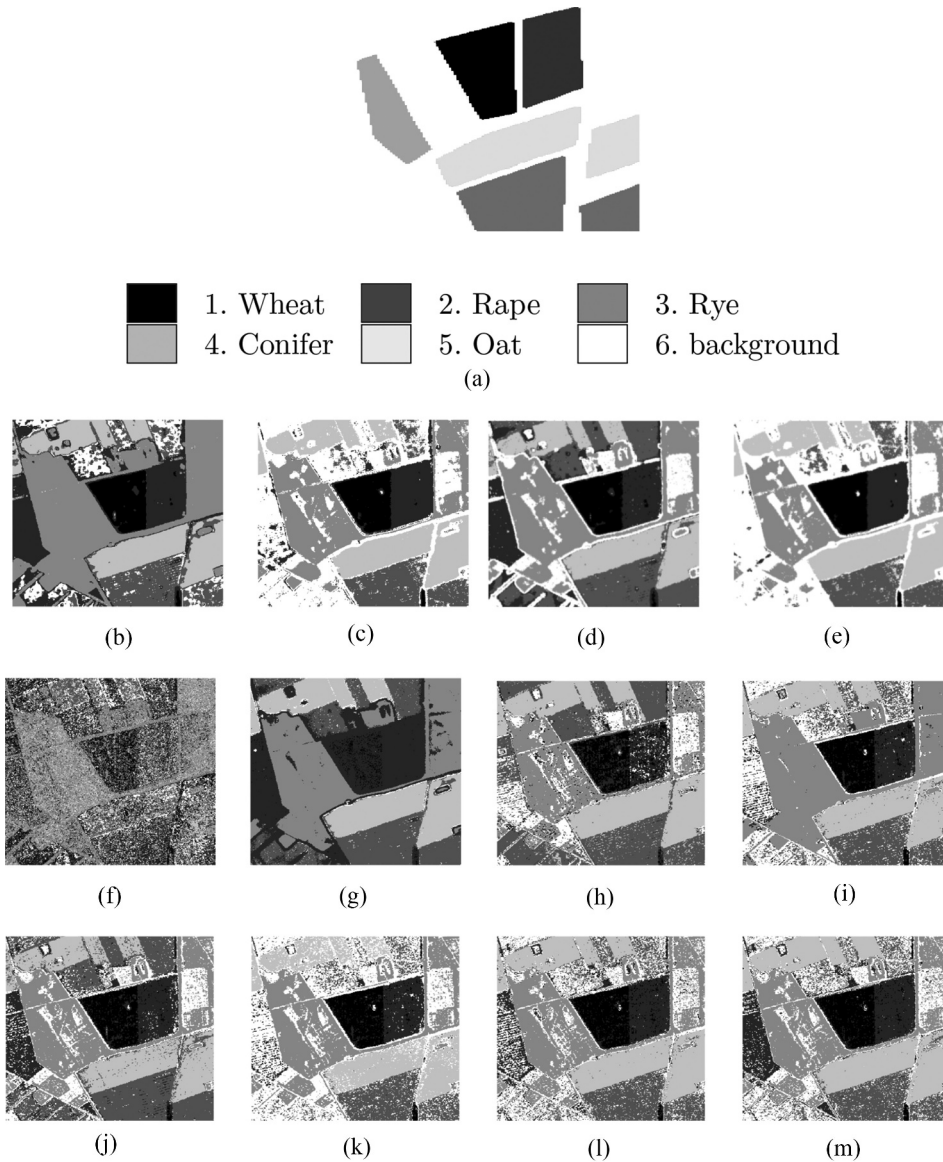


Figure 10. Classification results on the Foulum EMISAR image. (a) Ground-truth data of Foulum; (b) $D_{KL}^f(\theta_1||\theta_2) - 5 \times 5$; (c) $D_{KL}^f(\theta_1||\theta_2) - 7 \times 7$ - Box-Cox; (d) $D_{KL}^N(\theta_1||\theta_2) - 7 \times 7$; (e) $D_{KL}^N(\theta_1||\theta_2) - 7 \times 7$ - Box-Cox; (f) $D_{KL}^{St}(\theta_1||\theta_2) - 3 \times 3$; (g) $J^t(\theta_1||\theta_2) - 7 \times 7$ - Box-Cox; (h) LDA; (i) LDA - Box-Cox; (j) QDA; (k) QDA - Box-Cox; (l) KNN; (m) KNN - Box-Cox.

6. Conclusion

In this paper we have proposed a family with fourteen classifiers based on divergence measures between symmetric random vectors as well as hermitian random matrices. Specifically, we have considered Kullback-Leibler, Jeffreys, and Rényi discrepancies induced from normal, skew-normal, t -Student, skew- t , and scaled complex Wishart models. These measures were used by means of traditional Bayesian scheme like the core part of classifiers for trivariate SAR intensities (resulting from symmetric vector models) and for multilook PolSAR data (from the considered matrix model).

Proposals were assessed on both synthetic and actual data. For the former, we carried out Monte Carlo experiments on which generated data followed the $\mathcal{G}_p^0$ model that emulates real scenes with relief and multidimensional speckle noise having reciprocal gamma and (SCW) scaled complex Wishart models, respectively. Results have shown evidence in favour of the use of the optimized Rényi divergence between SCW models. We also performed two experiments with actual data. In general, for classification of vector SAR intensity data, the use of Box-Cox transformation have improved the classification rates in all cases. For the experiment with the AIRSAR image of San Francisco, the optimized Rényi divergence between Gaussian laws has obtained the best performance. For the experiment with the EMISAR image of Foulum, the classifier based on the KL measure between t -Student models achieved the best result, outperforming the classic LDA, QDA, KNN, and SVM classifiers as well as one in terms of the KL divergence for Gaussian models.

Future researches aim at adopting other multivariate models which include level curves more flexible than that we have used, considering asymmetric, nonparametric, and semi-parametric contexts.

Disclosure statement

No potential conflict of interest was reported by the authors.

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